Problem Set 1. B99602056 莊道茂

Ex1 10.

Take $f_1(x)=x$, $f_2(x)=x^2$, $f_3(x)=x^3$, $x\in[-1,1]$

By applying Gram-Schmidt process, we obtain that

$$\Rightarrow \psi_1 = \frac{|f_1\rangle}{\sqrt{\langle f_1|f_1\rangle}} = \frac{x}{\sqrt{\int_{-1}^{1} x^2 dx}} = \sqrt{\frac{3}{2}}\,x$$

↑ by using bra-ket notation.

$$\psi_2' = f_2(x) - |\psi_1\rangle\langle\psi_1|f_2\rangle = x^2 - \sqrt{\tfrac{3}{2}}x\left(\int_{-1}^{1}\sqrt{\tfrac{3}{2}}x\cdot x^2\,dx\right)$$ (odd func.)

$$= x^2 - \left(\sqrt{\tfrac{3}{2}}\right)^2 x\left[\tfrac{1}{4}x^4\Big|_{-1}^{1}\right]$$ (= 0)

$$= x^2$$

$$\Rightarrow \psi_2 = \frac{|\psi_2'\rangle}{\sqrt{\langle\psi_2'|\psi_2'\rangle}} = \frac{x^2}{\sqrt{\int_{-1}^{1}x^4dx}} = \sqrt{\frac{5}{2}}\,x^2$$

$$\psi_3' = f_3(x) - |\psi_1\rangle\langle\psi_1|f_3\rangle - |\psi_2\rangle\langle\psi_2|f_3\rangle$$

$$= x^3 - \sqrt{\tfrac{3}{2}}x\left(\int_{-1}^{1}\sqrt{\tfrac{3}{2}}x\cdot x^3\,dx\right) - \sqrt{\tfrac{5}{2}}x^2\left(\int_{-1}^{1}\sqrt{\tfrac{5}{2}}\underbrace{x^2\cdot x^3}_{\text{odd}}dx\right)$$ (= 0)

$$= x^3 - \sqrt{\tfrac{3}{2}}x\left(\sqrt{\tfrac{3}{2}}\cdot\tfrac{1}{5}(1+1)\right) - \sqrt{\tfrac{5}{2}}x^2\sqrt{\tfrac{5}{2}}\left(\tfrac{1}{6}x^6\Big|_{-1}^{1}\right)$$ (= 0)

$$= x^3 - \tfrac{3}{2}\cdot\tfrac{2}{5}\cdot x$$

$$= x^3 - \tfrac{3}{5}x$$

$$\Rightarrow \psi_3 = \frac{|\psi_3'\rangle}{\sqrt{\langle\psi_3'|\psi_3'\rangle}} = \frac{\left(x^3-\frac{3}{5}x\right)}{\sqrt{\int_{-1}^{1}\left(x^3-\frac{3}{5}x\right)^2dx}} = \frac{x^3-\frac{3}{5}x}{\sqrt{\int_{-1}^{1}\left(x^6-\frac{6}{5}x^4+\frac{9}{25}x^2\right)dx}}$$

$$= \sqrt{\frac{175}{8}}\left(x^3-\frac{3}{5}x\right)$$

$$= \frac{5}{2}\sqrt{\frac{7}{2}}\left(x^3-\frac{3}{5}x\right)$$

Therefore, the first 3 orthonormal functions are

$\psi_1(x)=\sqrt{\frac{3}{2}}\,x$, $\psi_2(x)=\sqrt{\frac{5}{2}}\,x^2$, $\psi_3(x)=\frac{5}{2}\sqrt{\frac{7}{2}}\left(x^3-\frac{3}{5}x\right)$, which is like a normal Legendre-polynomial.

Ex. 2.

$$\|\psi+\phi\| \overset{(1.3a)}{=} (\psi+\phi,\ \psi+\phi)^{1/2}$$

(in Hilbert-space, with norm: $\|\psi\| = \sqrt{(\psi,\psi)} \overset{\text{in bra-ket notation}}{=} \sqrt{\langle\psi|\psi\rangle}$)

$$\overset{(1.3a)}{=} (\psi+\phi,\psi)+(\psi+\phi,\phi)$$

$$\overset{(1.3c)}{=} (\psi,\psi+\phi)^* + (\phi,\psi+\phi)^*$$

$$\overset{(1.3a)\text{ again}}{=} \left[(\psi,\psi)+(\psi,\phi)\right]^* + \left[(\phi,\psi)+(\phi,\phi)\right]^*$$

$$= (\psi,\psi)^* + (\psi,\phi)^* + (\phi,\psi)^* + (\phi,\phi)^*$$

$$= (\psi,\psi) + (\phi,\psi) + (\psi,\phi) + (\phi,\phi)$$

$$= \|\psi\|^2 + \|\phi\|^2 + (\phi,\psi) + (\psi,\phi)$$

$$= \|\psi\|^2 + \|\phi\|^2 + \left[(\phi,\psi) + (\phi,\psi)^*\right]$$

$$= \|\psi\|^2 + \|\phi\|^2 + 2\,\mathrm{Re}\{(\phi,\psi)\}$$

(Re: real part. notation.)

(Since $|(\psi,\phi)| \geq \mathrm{Re}\{(\psi,\phi)\}$)

$$\leq \|\psi\|^2 + \|\phi\|^2 + 2\,|(\psi,\phi)|$$

$$\leq \underbrace{\|\psi\|^2 + \|\phi\|^2 + 2\,\|\psi\|\,\|\phi\|}_{\|}$$

$$(\|\psi\| + \|\phi\|)^2$$

Finally, $\Rightarrow$ $\boxed{\|\psi+\phi\|^2 \leq (\|\psi\|+\|\phi\|)^2}$

Ex. 3 10

By Ex. 2. we derived the following relation :

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + (x, y) + (y, x) \quad \text{—— ①}$$

thus, if we take y into $-y$, then we can obtain:

$$\|x+(-y)\|^2 = \|x\|^2 + \|(-y)\|^2 + (x, -y) + (-y, x)$$

$$\Rightarrow \|x-y\|^2 = \|x\|^2 + \|y\|^2 - (x, y) - (y, x) \quad \text{—— ②}$$

By (1.3b) in lecture notes.

Now, substituting ψ, ϕ into x and y in equation ① and ②.

$$\Rightarrow \begin{cases} \|\psi+\phi\|^2 = \|\psi\|^2 + \|\phi\|^2 + (\psi, \phi) + (\phi, \psi) & \text{—— ③} \\ \|\psi-\phi\|^2 = \|\psi\|^2 + \|\phi\|^2 - (\psi, \phi) - (\phi, \psi) & \text{—— ④} \end{cases}$$

③ plus ④

$$\Rightarrow \|\psi+\phi\|^2 + \|\psi-\phi\|^2 = 2\left(\|\psi\|^2 + \|\phi\|^2\right)$$

Ex. 4 10.

(Proof 1)

(i) If $(V, \|\cdot\|)$ is a normed vector space, the norm $\|\cdot\|$ induces a metric, thus there is a topological structure on V.

This metric is defined in the natural way, the distant between two vectors u, v is given by $\|u - v\|$.

(ii) ☆ Since all norms on a finite-dimensional vector space are equivalent from topological point of view (I will try to briefly prove this property!) although the resulting metric spaces need not be the same. (in proof 2)

(iii) And, since any Euclidean space is complete (I take this property for granted, since every Cauchy sequences in it are converge!!) Thus, it follows that all finite-dimensional vector space are Banach spaces.

(iv) Finally, in other words, by the definition of Banach space (which be defined as "complete normed vector space"), we can conclude that "every finite-dimensional (normed) vector space is (real / complex) Banach space and which is comple

depend on over real numbers / complex number field.

(Proof 2).

There are some important properties which I will show them in the following pages.

(i) Any two norms on a finite dimensional space are equivalent.

i.e. There exist some constants $c>0$, and $c'>0$ such that

$$c\|v\|_1 \leq \|v\|_2 \leq c'\|v\|_1.$$

for all vectors v, where $\|\cdot\|_1$ and $\|\cdot\|_2$ denote two different norm in this norm vector space $(V, \|\cdot\|_i)$. $\exists i \in I = \{1, 2, \cdots\}$.

(ii) Any finite-dimensional normed vector space is a Banach space (i.e. is complete)

Since (i) and (ii) are equivalent statement, in other words we can start from (i) to derive (ii) and vice versa. However, (i) => (ii) is need to use more technique in topology and real analysis, so I just write down the reference what I have found

→ "A course in functional analysis" by John. B. Conway P. 69 ~ 70.

And I will try to prove (ii) first, and then briefly to show how to derive (i) by topological method.

I will prove (i) in a special case, and extend to general case by using the theorem of "all norms (specific norm) are equivalent in finite-dimensional normed vector space."

Proof:

Suppose that $\mathbb{V}$ is a finite dimensional vector space over $\mathbb{F}$, with basis $\{v_1, \dots, v_r\}$ and norm $\|\cdot\|_0 : V \to \mathbb{R}$, defined for every $x_n = c_1 v_1 + c_2 v_2 + \cdots c_r v_r$, where $\mathbb{F}$ is a field and can be real numbers field, $\mathbb{R}$, or complex numbers field, $\mathbb{C}$. And, assume the

$$\|x_0\| = \left(\sum_{i=1}^{r} |c_i|^2 \right)^{1/2}.$$

We also suppose that $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{V}$. For each $n \in \mathbb{N}$, let $x_n = c_{n_1} v_1 + \cdots + c_{n_r} v_r$, where $c_{n_1}, \dots c_{n_r} \in \mathbb{F}$ are unique.

Since we can define a metric function

$$\rho : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$$

and equal to norm:

$$\rho_0(x_m, x_n) = \|x_m - x_n\|_0$$

$$= \left(\sum_{i=1}^{r} |c_{m_i} - c_{n_i}|^2 \right)^{1/2}$$

So that for every fixed $i = 1, \cdots, r$ we have

$$|c_{m_i} - c_{n_i}| \leq \rho_0(x_m, x_n)$$

and so $(c_{n_i})_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{F}$.

Since $\mathbb{F}$ is complete, it follows that for every fixed $i = 1, \cdots, r$ there exists $c_i \in \mathbb{F}$ such that $c_{n_i} \to c_i$ as $n \to \infty$,

therefore, given any $\epsilon > 0$, there exists N_i such that $|c_{n_i} - c_i| < \frac{\epsilon}{\sqrt{r}}$ whenever $n > N_i$. ...

So that let $x = c_1 v_1 + c_2 v_2 + \cdots + c_r v_r$.

Then we can obtain that

$$\rho_0(x_n, x) = \|x_n - x\|_0$$

$$= \left(\sum_{i=1}^{r} |c_{n_i} - c_i|^2 \right)^{1/2} < \epsilon$$

whenever $n > \max\{N_1, \cdots, N_r\}$.

It follows that $x_n \to x$ as $n \to \infty$. Hence $\mathbb{V}$ is complete.

Now, I will expend this result to any norm in finite-dimensional normed vector space.

Since we have a theorem : Every finite dimensional normed vector space over $\mathbb{F}$, $\mathbb{F}$ is complete, with respect

to the basis $\{V_1, V_2, \dots, V_r\}$. Then any norm $\|\cdot\| : V \to \mathbb{R}$ is equivalent to the norm $\|\cdot\|_0 : V \to \mathbb{R}$.

Thus, we complete the proof that in any finite-dimensional normed vector space, the space is complete, i.e. Banach-space.

real or complex ($\mathbb{R}$ or $\mathbb{C}$)
depends on which field the V is over on it

Ex 5. By the definition of the inner product in our Lecture note (1.3b) and property of vector space "(1.1h)", we can show this is valid.

Suppose $\phi = \underset{\sim}{0}$ and $a = \underset{\sim}{0}$ → number zero.
null vector, notation with capital zero!

(i) Then by (1.3b) we have $(\psi, a\phi) = a(\psi, \phi)$

$\Rightarrow$ right hand side $= a(\psi, \phi)$

$= 0(\psi, 0)$

$= 0$ — ①

left hand side: $(\psi, a\phi) = (\psi, 0)$ — ②
$\because a\phi = 0$ By (1.1h)

$\therefore$ l.h.s = r.h.s. (i.e. ① = ②)

Thus $(\psi, 0) = 0$.

(ii) $\because (\psi, a\psi) = a(\psi, \psi) = (a\psi, \psi) = (\underset{\sim}{0}, \psi)$ — ③
$a = 0$; null vector

and $(\psi, a\psi) = (\psi, 0)$ — ④

Therefore, by ③ = ④, we obtain that.

$(\psi, 0) = (0, \psi)$

and according (i), we conclude that.

$(\psi, 0) = (0, \psi) = 0$

Ex. 6 (i) Show that in $L^2[-1,1]$ space, norm can be defined as $\|f(x)\| = \sqrt{(f(x), f(x))}$.

(ii) And, also can be defined as

$$\|f(x)\| = \max\{|f(x)|,\ a \le x \le b\}.$$

(pf i) ① $\|f(x)\| = \sqrt{(f(x), f(x))} \ge 0$ ($\because \sqrt{a} \ge 0$, $a \in \mathbb{R}$)

$\|f(x)\| = 0$ iff $f(x)$ is null.

② $\|af(x)\| = \sqrt{(af(x), af(x))}$

$= \sqrt{a^2 (f(x), f(x))}$

$= |a| \sqrt{(f(x), f(x))}$

$= |a| \|f(x)\|$

③ $\|f_1(x) + f_2(x)\|$

$= \sqrt{(f_1(x) + f_2(x),\ f_1(x) + f_2(x))}$

$= \sqrt{(f_1(x), f_1(x)) + (f_1(x), f_2(x)) + (f_2(x), f_1(x)) + (f_2(x), f_2(x))}$

$\int_{-1}^{1} f_2 f_1 \, dx$

$\Rightarrow \|f_1(x) + f_2(x)\|^2 = (f_1(x), f_1(x)) + \underbrace{(f_1(x), f_2(x))}_{\int_{-1}^{1} f_1 f_2 \, dx} + (f_2(x), f_1(x)) + (f_2(x), f_2(x))$

$\le \|f_1(x)\|^2 + \|f_2(x)\|^2 + 2\int_{-1}^{1} |f_1 \| f_2| \, dx$

≤ 1

$(\|f_1(x)\| + \|f_2(x)\|)^2$

Because by the Schwarz inequality

$\int_{-1}^{1} |f_1(x)| |f_2(x)| \, dx \le \left(\int_{-1}^{1} |f_1(x)|^2 dx\right)^{1/2} \left(\int_{-1}^{1} |f_2(x)|^2 dx\right)^{1/2} = \|f_1(x)\| \|f_2($

Finally, $\|f_1\|^2 + \|f_2\|^2 + 2\int_{-1}^{1} |f_1||f_2|\,dx \le (\|f_1\| + \|f_2\|)^2$

that is, $\|f_1 + f_2\|^2 \le (\|f_1\| + \|f_2\|)^2$

$\Rightarrow \|f_1(x) + f_2(x)\| \le \|f_1(x)\| + \|f_2(x)\|$

Therefore, $\|f(x)\| = \sqrt{(f(x), f(x))}$ is a norm is $L^2(-1,1)$ space.

(Proof of (ii))

① since $|f(x)|$ always positive or larger/equal to zero, $|f(x)| \ge 0$,

($|f(x)| = 0$ iff $f(x) = 0$)

thus, $\max\{|f(x)|, a \le x \le b\} \ge 0$.

and then $\|f(x)\| \ge 0$. → condition ① is valid!

② $\|a'f(x)\| = \max\{|a'f(x)|, a \le x \le b\}$.

note that $a' \neq a$

$= \max\{|a'||f(x)|, a \le x \le b\}$

$= |a'| \cdot \max\{|f(x)|, a \le x \le b\}$

$= |a'| \cdot \|f(x)\|$ → Condition ② is valid!

③ $\|f_1(x) + f_2(x)\| = \max\{|f_1(x) + f_2(x)|, a \le x \le b\}$.

$$\Rightarrow \max\{|f_1(x)+f_2(x)|,\ x\in[a,b]\} \le \max_{x\in[a,b]}\{|f_1(x)|+|f_2(x)|\}$$

$$\le \max_{x\in[a,b]}\{|f_1(x)|\} + \max_{x\in[a,b]}\{|f_2(x)|\}$$

$$= \|f_1(x)\| + \|f_2(x)\|$$

i.e. $\|f_1(x)+f_2(x)\| \le \|f_1(x)\| + \|f_2(x)\|$.

It follows that, the condition ③ is also valid for the maximum norm : $\|f(x)\| = \max\{|f(x)|,\ a\le x\le b\}$

or write it in another notation to more briefly

$$\Rightarrow \|f(x)\| = \max_{x\in[a,b]}\{|f(x)|\}$$

which is a norm on $L^2(-1,1)$ space.